CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

3.2 Non-Uniform Motion

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

An egg is dropped from the top of a three-storey building. It falls through air until it reaches the ground

What remains constant throughout the fall?

A acceleration of the egg

B weight of the egg

C velocity of the egg

D air resistance on the egg

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Weight = mass × acceleration of free fall (g)
- Both the mass of the egg and the value of g stay constant during the fall

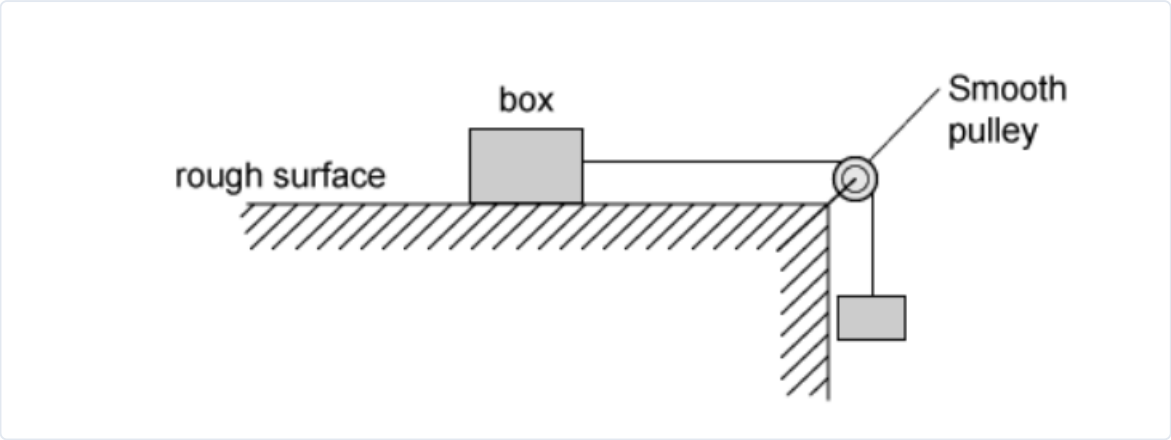
A is incorrect because the acceleration decreases as the magnitude of the air resistance increases

C is incorrect because the velocity increases until it reaches its terminal velocity

D is incorrect because air resistance increases as the velocity of the egg increases

Question 2 Easy [1 mark]

A box rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a mass at its other end.



When the box is released, what forces are acting on it?

A weight only

B friction and weight

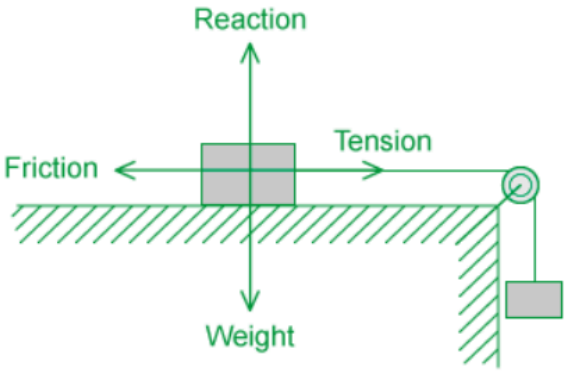
C friction, weight and reaction

D tension, friction, weight and reaction

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The weight of the mass below the pulley creates a tension in the string that pulls the box to the right
- Friction acts to the left, opposing the motion
- The weight of the box acts downwards
- Since the box is on a surface, there is an upward reaction force between the surface and the box

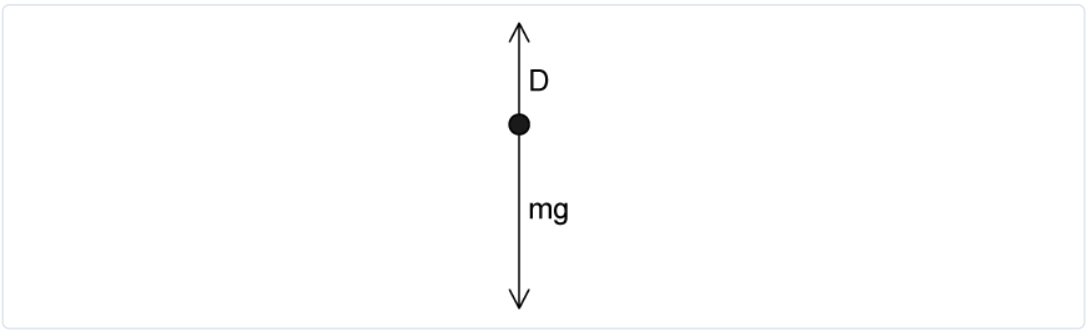

Reaction
Friction
Tension
Weight

Question 3 Easy [1 mark]

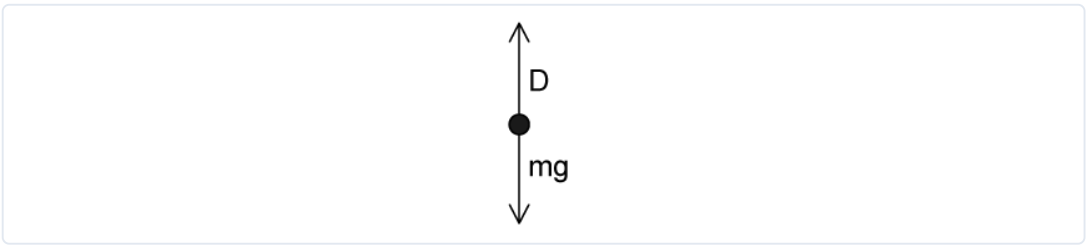
An object of weight *mg* falls through the air and experiences a drag force *D*.

Which of the following free-body diagrams shows the object when it has reached terminal velocity?

A



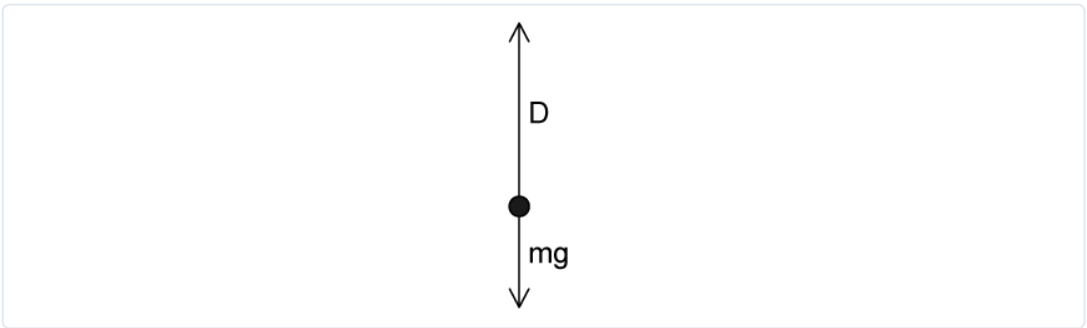
B



C



D



Mark Scheme & Explanation:

*The correct answer is **B** because:*

- When an object reaches terminal velocity, the drag force and the weight are equal

- The diagram in option **B** has equal force arrows for the weight and drag force
    - So the object is traveling at terminal velocity

Option **A** is incorrect because:

- The weight is greater than the drag force
- The object is accelerating downwards

Option **C** is incorrect because:

- The only force the object experiences is its weight
- The object is accelerating downwards

Option **D** is incorrect because:

- The drag force is greater than the weight
- The object is decelerating

Question 4 Easy [1 mark]

A skydiver has a weight of 800 N, and experiences a drag force of 450 N.

What is the magnitude and direction of the resultant force of the skydiver?

	Magnitude / N	Direction
A	1250	Downwards
B	350	Upwards
C	1250	Upwards
D	350	Downwards

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The free body diagram will look as follows



- The weight of the skydiver is greater than the drag force
  - Weight always acts downwards
  - The resultant force is therefore also downwards
- The magnitude of the resultant force will be the difference between the weight and the drag force

- 800-450=350 N

- This is option **D**

Question 5 Easy [1 mark]

A lorry is driving along a horizontal road and experiences a resistive force opposing the forward motion.

Assuming that air resistance is negligible, what is the name of the resistive force?

A Upthrust

B Weight

C Friction

D Normal contact force

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The wheels on the lorry are in contact with the road
- A frictional force from the road will oppose the motion of the lorry

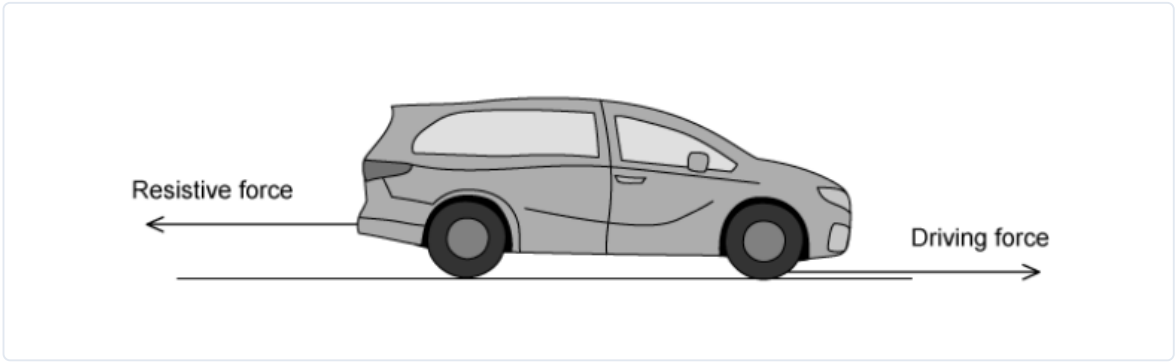
A is incorrect as upthrust opposes weight for a body in a fluid

B is incorrect as weight acts downwards and perpendicular to the lorry's direction of motion, so it will not resist the forward motion of the lorry.

D is incorrect as the normal contact force acts upwards and balances the weight, so there is no vertical acceleration.

Question 6 Medium [1 mark]

A car has a horizontal driving force of 2.0 kN acting on it and a resistive force a quarter of this size. It has a forward horizontal acceleration of 2.0 m s^{-2}



What is the mass of the car?

A 500 kg

B 750 kg

C 1250 kg

D 2000 kg

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Let's take the positive direction as the direction of the driving force and the negative direction as the direction of the resistive force
- The driving force is given as 2.0 kN = 2000 N
- The resistive force is $\frac{2000}{4} = 500 \text{ N}$
- The resultant force is F = 2000 – 500 = 1500 N
- Using Newton's second law:
  - $F = ma$
  - $m = \frac{F}{a} = \frac{1500}{2} = 750 \text{ kg}$

Question 7 Medium [1 mark]

A stone of mass *m* is dropped from a tall building. There is significant air resistance. The acceleration of free fall is *g*.

When the stone reaches its terminal velocity, which row is correct?

	Magnitude of the acceleration of the stone	Magnitude of the force of gravity on the stone	Magnitude of the force of air resistance on the stone
A	zero	*mg*	*mg*
B	*g*	*mg*	*mg*
C	*g*	zero	zero
D	zero	zero	zero

A

B

C

D

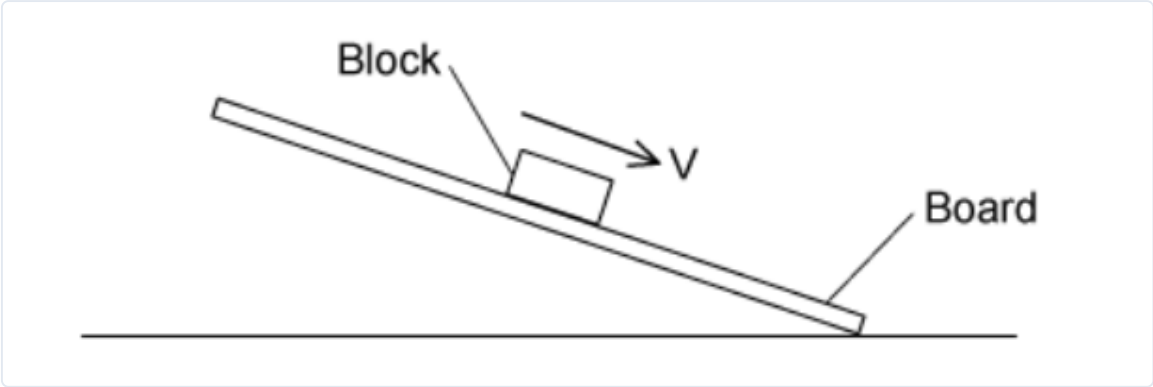
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- At terminal velocity, the speed is constant
- So, there is no acceleration meaning the resultant force on the stone is zero
- The weight of the stone (*mg*) remains unchanged throughout the duration of the fall
- So, for the resultant force to be zero, the magnitude of the air resistance should be the same as the weight (*mg*)

Question 8 Medium [1 mark]

A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at constant velocity *v*



Which row describes the forces acting on the block when sliding with constant velocity?

	Frictional force on block	Resultant force on block
A	down the plane	zero
B	down the plane	down the plane
C	up the plane	zero
D	up the plane	down the plane

A

B

C

D

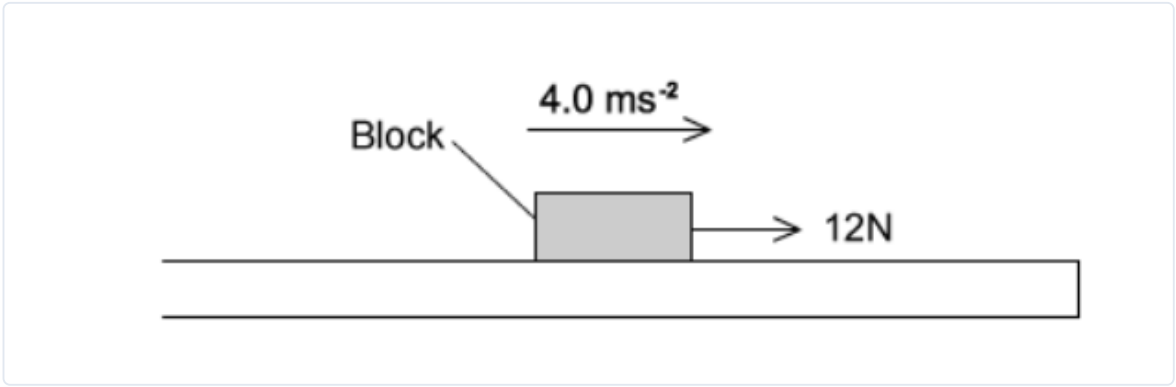
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The question tells us the block slides down the plane with constant velocity meaning there is no acceleration and hence the resultant force is zero
- Frictional forces oppose motion, and since the block moves down the plane, the friction force must act up the plane

Question 9 Medium [1 mark]

A block of mass 1.20 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2}



What is the magnitude of the frictional force acting on the block?

A 2.4 N

B 3.6 N

C 7.2 N

D 16.8 N

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Let's call the frictional force F_f
- Then, we can use $F = ma$ to equate the resultant force (= forward force – frictional force) with the mass and acceleration of the block
  - $F = 12 - F_f = 1.2 \times 4 = 4.8$
  - $12 - 4.8 = F_f$
  - $F_f = 7.2 \text{ N}$

Question 10 Medium [1 mark]

A boat travels forward in a straight line. The driving force of the boat is less than the water resistance acting on the boat.

Which statement is true?

A The boat is travelling at constant velocity

B The resultant force acts in the same direction as the motion

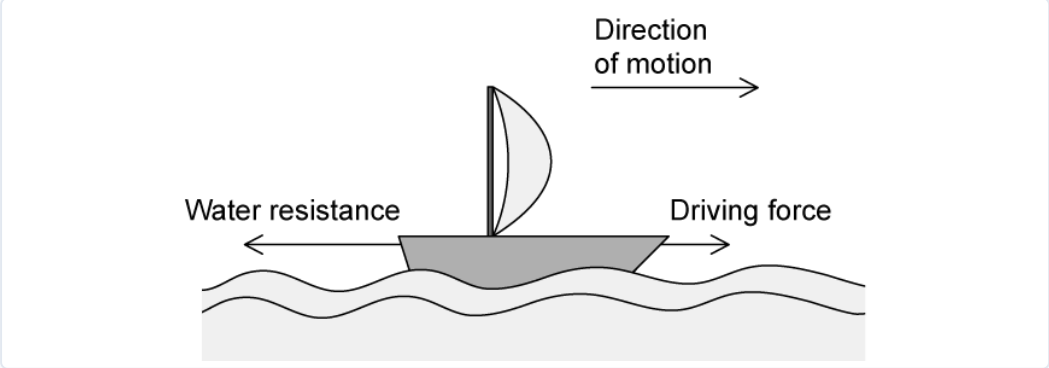
C The boat is decelerating

D The resultant force is equal to the driving force of the boat

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The free-body diagram of the boat is shown below
- The driving force acts in the same direction as the boat is travelling in



- Water resistance acts in the opposite direction to which the boat is travelling in
- Therefore, if the boat is moving forward but the driving force is less than the water resistance, it must be **decelerating**

Option **A** is incorrect because:

- For the boat to be travelling at constant velocity, the resultant force must be zero
    - As the driving force and resistive force are not equal, the boat is not travelling at constant velocity

Option **B** is incorrect because:

- The resistive force is greater than the driving force

- Water resistance acts in the opposite direction to which the boat is travelling in
- The resultant force is therefore in the same direction as the resistive force

Option **D** is incorrect because:

- The resultant force is equal to the difference between the driving force and the resistive force

Question 11 Hard [1 mark]

The three forces acting on a hot air balloon as it moves vertically are its weight, the force due to air resistance and upthrust.

The hot air balloon descends vertically at constant speed. The force of air resistance on the balloon is F.

What weight of material must be released from the balloon so that it ascends vertically at the same constant speed?

A $\frac{1}{2}F$

B F

C $2F$

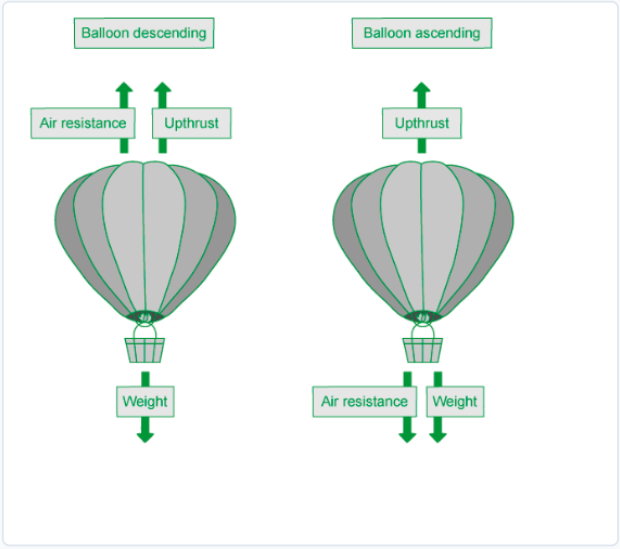
D $4F$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- As the balloon ascends, the forces acting are as follows:
  - The weight W always acts downwards and has a constant value
  - The upthrust U acts upwards and depends on the volume of air displaced by the balloon
  - Air resistance F opposes the direction of motion, so if the balloon is ascending, the air resistance acts downwards and changes depending on the speed
- When the balloon is descending at a constant speed, air resistance = F
- Constant speed means the acceleration, and hence, the resultant force = 0
- Therefore, upward forces = downward forces
  - Upward forces = $F + U$
  - Downward forces = W
  - $W = F + U$ **(1)**
- When the balloon is ascending at the same constant speed, a weight X is released, so the new weight is $W - X$
  - Upward forces = U
  - Downward forces = $W - X + F$
  - $F + W - X = U$ **(2)**
- Substituting **(1)** into **(2)**

- $F + F + U - X = U$
- $2F - X = 0$
- $X = 2F$



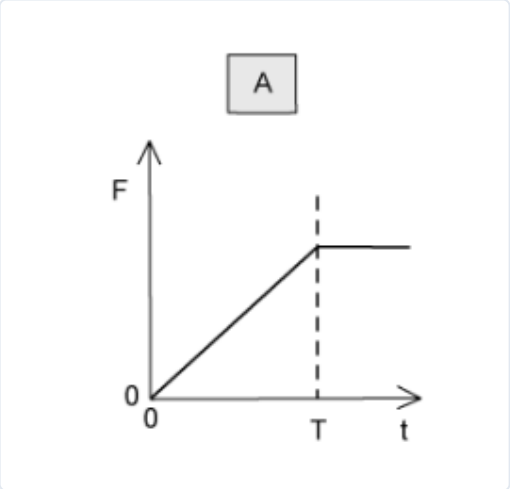
Question 12 Hard [1 mark]

A skydiver falls vertically from a stationary balloon.

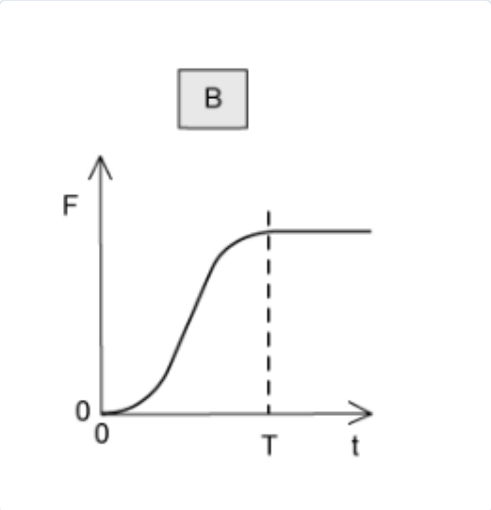
She leaves the balloon at time $t = 0$. At time $t = T$, she reaches terminal velocity. Beyond the time shown in the graphs, she opens her parachute

Which graph shows the variation with time t of the force F due to air resistance?

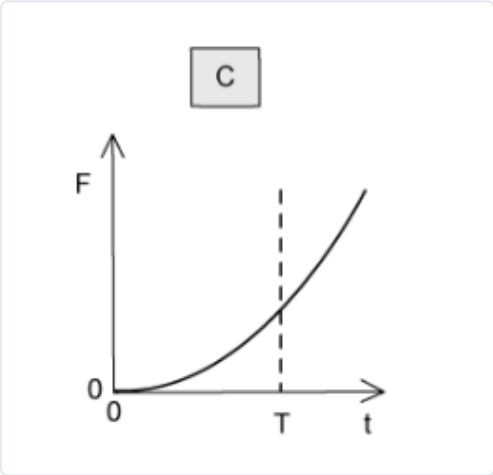
A



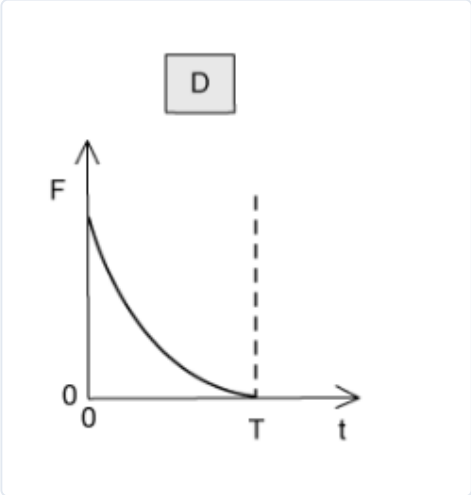
B



C



D

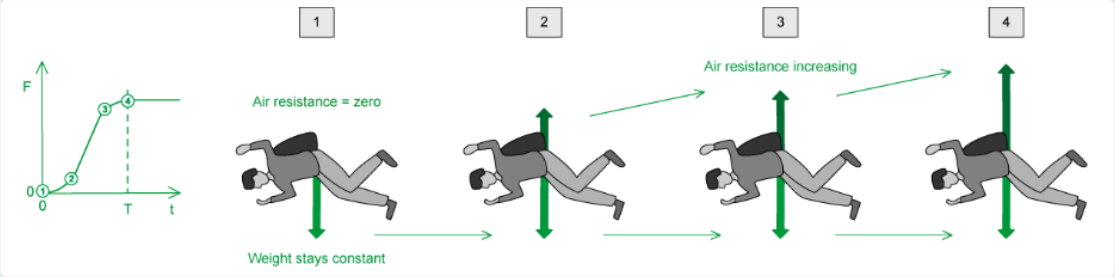


Mark Scheme & Explanation:

*The correct answer is **B** because:*

- **(1)** At $t = 0$, the skydiver's speed is initially zero, so air resistance is negligible
  - This means the graph must start from zero as $F \approx 0$
- **(2)** In the first few seconds, the increase in air resistance is relatively small, as the skydiver's speed is initially low and air resistance increases with speed
  - This means the increase in air resistance F is initially small
- **(3)** The skydiver continues to accelerate due to gravity, so as the speed increases, the air resistance increases too
  - This means air resistance F continues to increase until $t = T$
- **(4)** At $t = T$, the skydiver reaches terminal velocity, so air resistance and weight are equal. This means there is no resultant force, so the skydiver is no longer accelerating, so her speed remains constant (until she opens her parachute)

- This means the graph plateaus as air resistance F remains constant



A is incorrect because the rate of increase of air resistance is not constant so the gradient of the graph cannot be constant

C is incorrect because the air resistance cannot keep increasing after time T as this would imply it becomes larger than weight so the resultant force would increase upwards

D is incorrect because at time t = 0, the skydiver is stationary so both the speed and air resistance start from zero

Question 13 Hard [1 mark]

Two skydivers, A and B, experience drag forces D_1 and D_2 respectively. Skydiver A has mass m and reaches terminal velocity v_1. Skydiver B has mass $2m$ and reaches terminal velocity v_2.

Given that $D \propto v^2$, what is the value of $\frac{v_1}{v_2}$?

A $\frac{1}{2}$

B $\frac{1}{4}$

C 2

D $\frac{1}{\sqrt{2}}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- At terminal velocity the drag force is equal to the weight
  - $D = mg$ therefore
  - $mg \propto v^2$
- The terminal velocity of Skydivers A and B can be written respectively as
  - $mg \propto v_1^2$
  - $2mg \propto v_2^2$
- Therefore the ratio of v_1 to v_2 is
  - $\frac{mg}{2mg} = \frac{v_1^2}{v_2^2}$
  - $\frac{1}{2} = \frac{v_1^2}{v_2^2}$
  - $\frac{1}{\sqrt{2}} = \frac{v_1}{v_2}$
- This is option **D**

Question 14 Hard [1 mark]

A car of mass m starts from rest and accelerates uniformly, reaching a velocity of v over distance s. The car experiences a frictional force of $\frac{mg}{2}$.

What is the driving force of the car?

A $\frac{m}{2}\frac{v^2}{s} + g$

B $m\frac{v^2}{s} + \frac{W}{2}$

C $\frac{m}{2}\frac{v^2}{s} - g$

D $m\frac{v^2}{s} - g$

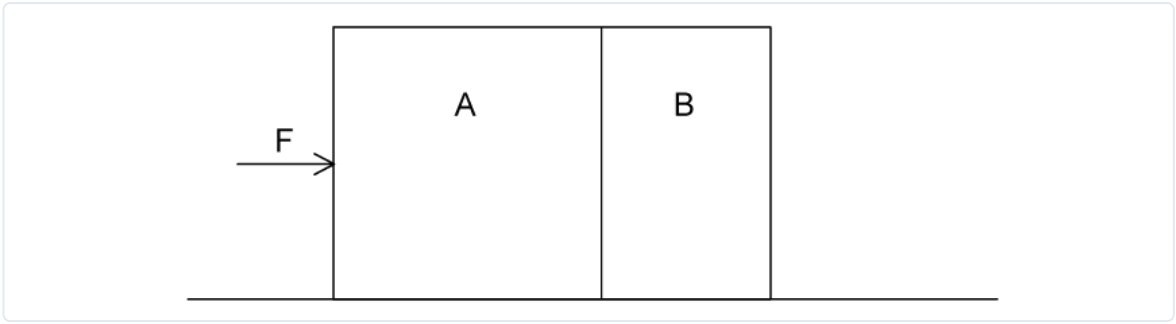
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The acceleration of the car is
  - $v^2 = u^2 + 2as$, where u = 0, as the car starts from rest
  - $v^2 = 2as$
  - $a = \frac{v^2}{2s}$
- The resultant force acting on the car is
  - $F = ma$
  - $F = F_D - F_f$, where F_D is the driving force and F_f is the frictional force
  - $ma = F_D - F_f$
- The driving force of the car is
  - $F_D = ma + F_f$
- Substituting in the expressions for acceleration and frictional force
  - $F_D = \frac{mv^2}{2s} + \frac{mg}{2}$
  - $F_D = \frac{m}{2}\frac{v^2}{s} + g$
- This is option **A**

Question 15 Hard [1 mark]

The diagram shows two blocks, A and B, that have masses of $3m$ and $2m$ respectively. A force F is applied to Block A, and both blocks accelerate along a horizontal surface. Block B experiences a drag force of $\frac{F}{5}$.



What is the resultant force of Block A on Block B?

A $\frac{1}{5}F$

B $\frac{4}{25}F$

C $\frac{8}{25}F$

D $\frac{2}{5}F$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The equation for the resultant force is
  - $F = ma$
- Both boxes can be considered to move as one object with
  - a total mass of $3m + 2m = 5m$
  - an acceleration of a
  - a net resultant force of $F_{net} = 5ma$
- As the boxes move together, the forces acting on A and B separately can be considered to act on the object as a whole
- The forces acting are F forward and $\frac{1}{5}F$ backward, so the net resultant force on the object is:
  - $F_{net} = F - \frac{1}{5}F = \frac{4}{5}F$
- The acceleration of A and B can be found by equating the expressions for net resultant force:

- $5ma = \frac{4}{5}F$
- $a = \frac{4F}{25m}$

- Therefore the force exerted by Block A on Block B is equal to
  - $F_{AB} = 2m \times \frac{4F}{25m}$
  - $F_{AB} = \frac{8}{25}F$
- This is option **C**